CAMBRIDGE INTERNATIONAL EXAMINATIONS

International General Certificate of Secondary Education

MARK SCHEME for the October/November 2013 series

0620 CHEMISTRY

0620/22 Paper 2 (Core Theory), maximum raw mark 80

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1 **(a)** **(i)** ammonia [1]

(ii) methane [1]

(iii) ammonium chloride [1]

(iv) water [1]

(v) calcium carbonate [1]

(vi) copper(II) sulfate [1]

(b) substance which contains two (or more) elements chemically combined (or bonded) / two different atoms bonded (or combined or joined) / different atoms bonded [1]
ALLOW: a substance containing two (or more) elements which cannot be separated by physical means

(c) CO_2 on right [1]
$2(O_2)$ [1]
NOTE: second mark dependent on first mark

[Total: 9]

2 **(a)** zinc → magnesium → calcium → sodium [2]
NOTE: 1 mark if one pair incorrectly placed / metals in reverse order

(b) magnesium chloride [1]
hydrogen [1]

(c) ion [1]

(d) 1 electron in outer shell [1]
8 electrons in middle shell [1]

(e) **(i)** correct method of collection i.e. upturned measuring cylinder over water or (gas) syringe [1]
workable apparatus and closed system [1]
flask or test tube labelled AND measuring cylinder or syringe labelled [1]
ALLOW: flask / test tube / syringe / measuring cylinder not joined up

(ii) Any three of: [3]

increase concentration (of hydrochloric acid) / use concentrated acid
increase temperature / heat up reaction
use smaller lumps of zinc /
add a catalyst

[Total: 13]

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3 **(a)** distillation [1]
ALLOW: (fractional) distillation

(b) (round-bottomed) flask [1]
thermometer [1]
condenser [1]
ALLOW: condensing tube

(c) 1 mark each:

lower
boils
condenses [3]

(d) **(i)** chloride / Cl^- [1]

(ii) K^+ / potassium [1]

(iii) Mg^{2+} [1]
SO_4^{2-} [1]

[Total: 11]

4 **(a)** 1 mark each: [4]

poly(ethene) → it has a very long chain
ethene → it decolourises bromine water
methane → it is the main constituent of natural gas
ethanoic acid → it contains a –COOH functional group

(b) **(i)** substance containing carbon and hydrogen <u>only</u> [1]

(ii) it has a double bond [1]

(c) monomers [1]

(d) **(i)** addition of oxygen / increase in oxidation number / loss of electrons [1]
ALLOW: removal of hydrogen

(ii) glucose (on left) [1]
ALLOW: sugar
carbon dioxide (on right) [1]

[Total: 10]

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5 **(a)** Any three of: [3]

alloy is a mixture / alloy is a combination of metal with another metal / of metals / of a metal with a non-metal
IGNORE: mixed with another substance /
alloying alters property of metal /
makes metal stronger /
makes metal more corrosion resistant /
makes metal harder /
ALLOW: reduces rusting ONLY if iron / steel mentioned
IGNORE: lasts longer / durable
ALLOW: answers from diagram
ALLOW:: higher level answers e.g. layers in metals slide over each other easily / layers in alloy do not slide as easily

(b) **(i)** 1 mark each: [2]
3rd box and 5th box ticked

(ii) 1 mark for method and 1 mark for why it works: [2]
painting / tinning / galvanising / covering with plastic / chromium / greasing / (electro)plating (1)
IGNORE: covering / coating (unqualified)
prevents air (or oxygen) and water coming into contact with iron (1)
OR
galvanising / coating with zinc / putting block of named reactive metal on surface (1)
metal reacts instead of iron / metal more reactive than iron (1)
ALLOW: sacrificial protection

(c) **(i)** substance which speeds up reaction / increases rate of reaction [1]

(ii) (damp) red litmus paper [1]
ALLOW: universal indicator
turns blue [1]
ALLOW: (concentrated) hydrochloric acid (1) white fumes (1)

(iii) Any two of: [2]

replacement of nitrogen / nitrates / potassium / phosphorus (taken up by plants)
plants take up nitrogen / potassium / phosphorus / nitrates from soil / nitrogen (or potassium or phosphorus) needed by plants
(fertiliser) adds extra nitrogen / potassium / phosphorus / nitrates (to replace this)
increase plant growth / plants grow better / plants grow faster / better yield
IGNORE: for plant growth / for healthy plants
make more (plant) protein

[Total: 12]

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6 **(a)** Any three of: [3]

evaporates or evaporation (from garlic) / idea of change from liquid to gas /
movement of particles / atoms / molecules / diffusion / particles (in garlic smell) collide (with air particles) /
spreading out or mixing up of particles / atoms / molecules /
random / disorderly (movement of particles / atoms / molecules) /
ALLOW: particles move from high(er) to low(er) concentration

(b) **(i)** $C_6H_{10}S_2$ [1]

(ii) (one) more sulfur atom in A / B has 1 sulfur atoms but A has 2 [1]
same number of C and H atoms / molecule otherwise the same / [1]

(c) **(i)** 18 [1]

(ii) atoms of same element with different number of neutrons / atoms with same number of protons and different numbers of neutrons / atoms differing only in number of neutrons / elements with same number of protons and different number of neutrons / elements with same proton number but different nucleon (or mass) number [1]
number of protons + neutrons (in an atom) [1]

(iii) coal; oxidised; dioxide; water; [4]

(iv) pits surface/ idea of (chemical) weathering / (chemical) erosion [1]
ALLOW: damages building / eats away the building / dissolves building / wears away the building / surface disintegrates / surface crumbles
IGNORE: destroys buildings / cracks the building / corrosion
acid (rain) reacts with carbonate / limestone / neutralisation [1]
REJECT: burns carbonate / melts carbonate

[Total: 15]

7 **(a)** **(i)** (limestone added): A [1]
(waste gases exit): B [1]

(ii) CO_2 [1]

(iii) 15 (g) [1]

(b) **(i)** harder / slower to decompose down Group / (ease) decreases down Group /
easier to decompose up Group / ease increases up Group / thermal stability increases down Group / thermal stability decreases up Group [1]
ALLOW: the more reactive the <u>metal</u>, the higher the decomposition temperature

(ii) **ALLOW**: values from 1000 to 2000 (^oC) (actual = 1360 ^oC) [1]

(c) **(i)** neutralise acidic soils / neutralise acidic lakes / making mortar / making calcium hydroxide / making limewater / whitewash [1]

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(ii) basic [1]
IGNORE: alkali / metal

(iii) 56 [1]

(d) (calcium) too reactive / (calcium) above carbon in reactivity series [1]
ALLOW: very reactive / high reactivity / more reactive than carbon

[Total: 10]

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